

Multiple Choice (10 marks)

1. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above
2. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.
3. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
 - (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs
4. In Python 3, what is ['a', 'Chemistry', 0, 1][-3]?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
5. In Python 3, what is ['a', 'Chemistry', 0, 1][1]?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1

True / False (10 marks)

Answer True or False

6. True or False: The function $O(\log \log N)$ grows the slowest among common complexity classes.
7. True or False: The Python expression `'abc'[::-1]` results in an error due to invalid syntax.
8. True or False: The `sort()` method can be directly called on a string to sort its characters in Python.
9. True or False: In Python 3, the expression `'a' + 'ab'` evaluates to `'aab'`.
10. True or False: In Python 3, the expression `x >> 1`, where `x = 8`, evaluates to 4 due to the right shift operation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of floor division in Python 3, explaining its significance and how it differs from regular division. Provide examples to illustrate your points.
12. Discuss the implications of a large Java program passing extensive testing without errors. What conclusions can be drawn about potential bugs, and why is thorough testing not always sufficient?

End of Paper